

CSEN 605: Digital System Design
*Winter 2025***Assignment and Project Document*****Part 1 Homework Assignment***

You are going to develop a small project based on FPGA to design a digital clock that displays the time in the format of hours, minutes and seconds.

1. Design a clock with a frequency of 50 GHZ.
2. Assume that every clock cycle corresponds to one second.
3. Also every minute is composed of 60 seconds, and every hour is composed of 60 minutes.
4. The seconds, minutes, and hours should be of type integer when defined in your code.
5. A reset to the clock can be applied, the type of the reset is synchronous active low.
6. You are required to write both the VHDL code and a testbench to simulate the clock.
7. You must apply the reset at least once in the testbench, and in the next clock cycle, the clock should continue normally.
8. You need to test and simulate your project using ModelSim and submit a screenshot for the simulation waveform (timing diagram).

Part 2 Project Description

You are going to develop a project based on FPGA using the board provided to you (or optionally your own) and writing VHDL code.

The material in the course does not fully exploit the capabilities of the FPGA board so you must do so yourself by consulting the user manual as well as other sources. Some examples include Arduino connector with ADC, push buttons, seven-segment displays, an accelerometer, and a VGA output.

Deliver your project by presenting a demo (A 2 to 3 min video in English that clearly shows the project and how it works), in addition to a small report that outlines the idea, the set of sensors and parts you used, the implementation (codes, circuits, pin assignment) and the results. Moreover, A real model representing your application as much as possible should be implemented and presented.

Before you start your physical design of the project, use ModelSim to simulate/test your system and make sure it is working well.

You can buy many of the external devices from the local market at:

- Future Electronics (<https://store.fut-electronics.com/>)
 - Address: El-Saray Street, Abdo Basha, Abasia (in front of the Faculty of Engineering, Ain Shams University)
- RAM (<https://www.ram-e-shop.com/?srsltid=AfmBOoouYTSh-H2qqETMV5gS85BqjO6Bj6XlIgENnJms5Xie0awS7ow9Q>)
 - Address: 32 El Falaky St., Bab El Louk, El Tahrir, Cairo, Egypt.

You'll be divided into **Two Batches**:

Batch 1: From November 14 to November 28

Evaluations for Batch 1: From December 2 to December 3

Batch 2: From December 5 to December 19

Evaluations for Batch 2: From December 20 to December 21

Submission Guidelines:

1. Submit the whole project including all the source code files.
2. A video of the simulation of your system on ModelSim.
3. A report documenting the whole project idea, inputs and outputs, pin assignment, results and brief explanation of your code.
4. Video (2-3 min) showing a demo of the project running.
5. Notepad with your names, IDs, tutorial numbers, and emails.
6. Submit a folder for the assignment solution project including all the source code files, in addition to a screenshot for the simulation waveform timing diagram.
7. Upload all the above material as a single zip file to Google Drive and make the link shareable.
8. Rename your zip folder with the following format (TEAMNUMBER_BATCHNUMBER).
9. Prepare a drive link containing the zipped folder to be submitted through a Google form before the deadline (the form will be announced later for each batch).

Things you should consider

- Availability of the components (Project description has links to some stores and their contacts)
- Group formation: exactly 7 members per team or you'll be randomly assigned.
- The contribution of each member should be clear so think of how you will divide the tasks.
- Evaluation is on campus. You need to present the project to the instructor/TA and be prepared for questions and/or modifications. All group members must be present.

NOTE THAT:

- 1. NO EXTENSIONS BY ANY MEANS TO ANY BATCH UNDER ANY EXCUSE WILL BE POSSIBLE. KINDLY STICK TO THE DATES STATED ABOVE.**
- 2. Sensors and actuators cannot be replaced by switch, button, buzzer, or LEDs, etc. It can only be used as additional devices; i.e., the project shouldn't be built upon a button and LED for example. It should be a REAL SENSOR such as (IR, proximity..etc) and the MOVING PARTS could be (motor, water pump..etc).**

Ideas

(You need to choose ONE of the following ideas)

A. PROJECT: "Smart Study Desk"

You are required to design a smart study desk such that it can sense the presence of the user and enhance the light intensity of the room if needed. Once the user approaches the desk he/she shall be able to grab what he/she needs from the drawer of the desk. Finally, the desk should record and display how much time the user used the desk.

Tasks:

1. The system needs to sense if a person approaches it.
2. If a person approaches the desk, the light intensity of the room shall be checked.
3. If the light intensity of the room is low, the lamp shall be turned on.
4. Unconditionally, if someone approaches the desk, the drawer shall open for 10 seconds and close afterward.
5. The user shall be able to open the desk manually at any time by pressing a button.
 - In this case, the drawer can be closed manually by pressing a button, or automatically after 10 seconds pass from opening it.
6. The time sat by the person on the desk shall be displayed in real-time on the 7 SEGMENT DISPLAY (in 2-digit seconds), and if the person leaves the time shall freeze.

B. PROJECT: “Smart Parking System”

A special place has a special type of garage, The garage has two modes autonomous and manual that determines how it opens and closes, and the number of empty spots are kept track of.

Tasks:

1. Initially, the number of empty spots is eight.
2. The mode (autonomous or manual) is set by a switch.
3. Entry to Garage:
 - a. Autonomous Mode:
 - The gate shall open automatically when a car gets near to it.
 - Then when the car passes the gate, it shall close automatically also, and the number of spots shall decrement by 1.
 - b. Manual Mode:
 - The gate shall open after pressing a button
 - The gate is close after pressing the button again and the count of spots will decrement by 1.
4. The count of the spots will be displayed on the 7-segment display.
5. A reset button shall also reset the system to the start (empty spots go back to 8)

C. PROJECT: “Walking Stick”

You are required to design an assistive walking stick for visually impaired individuals. The stick can be turned ON or OFF by the user and has multiple safety and guidance features. Its primary function is to detect obstacles in front of the user through an obstacle avoidance sensor, and when an obstacle is present, it alerts the user through vibration. In addition, if the stick is accidentally dropped from the user's hand, it automatically emits a beeping sound until it is retrieved.

Tasks:

1. Implement a switch/button (either through the FPGA or an external input) to allow the user to activate or deactivate the stick.
2. Use an obstacle detection sensor of your choice to trigger the vibration motor and blinking warning LED whenever an obstacle is detected.
3. Employ a touch sensor or equivalent to identify if the stick has been dropped, and if so, activate a sound device (e.g., buzzer) to continuously beep until the stick is recovered, at the same time a blinking led.
4. Add a status LED to indicate when the stick is active/inactive.
5. Program the 7-segment display on the FPGA to display the time.

D. PROJECT: “Smart Car”

You have a smart self-driving car that follows a straight lane. Whenever it goes out of the lane, a warning message should be displayed on the screen.

The car should also stop when it detects an obstacle in front of it. In addition, the car should detect if there is any gas leakage.

If gas leakage is detected, a warning sound should be played, and the car should stop.

Tasks:

1. Initially, The Car should be moving using a Moving Part.
2. The Car should follow a lane using a Sensor of your choice.
 - a. If the Car drifted outside its designated lane, a warning message “ALERT” should appear on the 7 SEGMENT DISPLAY.
3. The Car should detect if there is any Obstacle in front of it using a Sensor of your choice.
 - a. If the Car detected an Obstacle, The Car should stop.
4. The Car should detect if there is any Gas Leakage using a Sensor of your choice.
5. If The Car detects Gas Leakage, it should Stop, turn on a Red LED, and produce a Warning Sound using a sound device of your choice.

E. PROJECT: “Conveyer Belt System”

You have an automated conveyor belt used for transporting products. The belt remains stationary until a product is placed on it.

When a product is detected, the conveyor starts moving. If any obstacle or human is detected in front of the belt, the system should immediately stop and produce a warning sound.

The system should also keep count of the number of products that pass a specific point on the belt, displaying the count on the 7-segment display. A push button is used to reset the product counter to zero.

Tasks:

1. Initially, the Conveyor Belt should be stopped.
2. The Conveyor Belt should start moving when a product is placed on it using a Sensor of your choice.
 - a. If no product is detected, the Conveyor Belt should remain stopped.
3. The system should detect any Obstacle or Human in front of the belt using a Sensor of your choice.
 - a. If an Obstacle is detected, the Conveyor Belt should stop, and a Warning Sound should be played using a sound device of your choice.
4. The system should detect when a Product passes a specific point on the belt using a Sensor of your choice.
 - a. Each time a product passes, the 7-Segment Display should increment by 1 to show the total number of products.
5. An External Push Button should be used to reset the 7-Segment Display count back to 0.

F. PROJECT: “Minercraft”

Steve wants to reach diamonds, but his golden pickaxe keeps breaking before reaching them. So, he decided to build an automatic miner that uses low material and keeps digging until it reaches diamonds.

Tasks:

1. A sensor starts the miner’s engine when it detects any block. The miner continues digging until it finds diamonds. When it does, it stops, lights a yellow LED, and activates a buzzer to notify Steve.
2. To prevent damage, the miner must stop immediately upon detecting diamonds.
3. A seven-segment display shows how many diamonds the miner has collected
4. Steve can perform an emergency stop using a button.
5. Three LEDs indicate operation status:
 - 1) Mining (green led)
 - 2) Stopped (found diamond (yellow led))
 - 3) Emergency stop (red led)

Note: the buzzer in task 1 should stop after 5 seconds. Use any type of miners you want it doesn't have to be a drill.

Bonus: Add a chassis so the miner moves like a car — it moves forward only when blocks are detected and keeps mining until it reaches the diamond.

G. PROJECT “Smart Arcade Ball Shooter”

You are required to design a mini arcade ball shooter game using FPGA. The system allows a player to interact with it by firing balls toward a target. The game keeps track of the player’s score and provides visual and audio feedback to indicate hits or misses. A reset button is included to start a new game, and the 7-segment display is used to show both the current score and the remaining game time.

Tasks:

1. Implement a push button or touch sensor as the shoot trigger, activating a motor or solenoid to launch the ball toward the target.
2. Use a sensor to detect if the ball hits the target.
3. If hit, activate the green LED briefly and increment the score.
4. If missed, blink the red LED and activate the buzzer briefly.
5. Program a reset button to start a new game and reset the score and timer.
6. Display the current score on three 7-segment displays.
7. Display the remaining game time (fixed period, e.g., 60 seconds) on another three 7-segment displays, counting down in seconds.

Bonus: Store the highest score in registers to display the top score.

H. PROJECT: “StarPull”

Imagine pulling a character from a gacha game — each pull gives a random number that determines the character’s rarity. As the number of pulls increases (up to a maximum, e.g., 10), the chance of getting a higher-rarity character also increases by reducing the possible random integers to choose from.

Tasks:

1. A touch sensor triggers a pull.
2. A seven-segment display shows the current number of pulls.
Three colored LEDs indicate the rarity of the pulled character:
 - a. Blue for 1-star
 - b. Green for 2-star
 - c. Red for 3-star
3. A buzzer sounds when you win the highest-rarity (3-star) character.
Track the total time spent on the game. If the playtime exceeds a certain limit, a motor with a red flag activates to remind the gamer to stop playing.
4. Creative random number generation

Note: Increase the probability of higher-rarity pulls creatively with each attempt, without using floating-point numbers or division. This could be made by reducing the number of random integers to pull from.

Only three character types (1-, 2-, and 3-star) are needed; once the max pity count is reached, a 3-star is guaranteed.

Bonus: Use VGA output to display an animation or image of the pulled character on a screen / Add a microphone or speaker that plays music when pulling a 3-star character.

“BONUS”

- If you added an extra feature using an ANALOG SENSOR that requires an extra microcontroller connection (for example: Arduino Board) (or FPGA-based ADC module/code), it will be counted as a bonus to any of the ideas listed above. You must use the microcontroller for ONLY reading the sensor values; however, any output must be controlled through the FPGA.
- You can also have a bonus if you designed a creative model/prototype for your project.

Good Luck :)